

Jasper Sands

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RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for qubit characterization and control.

EDUCATION

Columbia University, New York, NY Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning.

Washington University in St. Louis, St. Louis, MO Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

Coursework: Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison · Parallel & Concurrent Programming · Computer Security · Reverse Engineering & Malware Analysis · Econometrics · Game Theory.

TECHNICAL REPORTS

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* Columbia University, 2025. jaspersands.com/assets/practical-llm-based-lossless-compression.pdf

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* Sep 2026 – May 2027

- Automating silicon spin qubit characterization under Dr. Nard Dumoulin Stuyck, targeting the manual tuning and interpretation of charge measurements that currently bottleneck semiconductor quantum hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Evaluating the automated analysis against how an expert reads the same device by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Desdr Open Insurance Toolkit, Columbia University · *Graduate Researcher* Sep 2025 – Dec 2025

- Built climate-risk and actuarial features from SMS survey data under severe sparsity with Dr. Eugene Wu, modeling expected agricultural losses to support micro-insurance program design.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian by grounding outputs in ontology and molecular graph data.

Foster System Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation. Measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XXXX surface codes with real-time error injection and syndrome visualization, running exact minimum-weight perfect matching and Union-Find decoding entirely client-side.
- Finite-size scaling estimates thresholds of 3.2% with MWPM, 2.7% with Union-Find, 14.7% at code capacity.

Q-Search: proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian hidden subgroup, dihedral coset, and linear code equivalence problems, pairing frontier theoretical reasoning with agentic engineering.
- Formal proof gates test each proposed mechanism against 1,200+ classical baselines across 720+ theorem modules, with negative results permanently indexed so the engine does not re-propose a dead end.

Noise-Aware Adiabatic Simulation: transverse-field Ising model · *Qiskit* IBM Heron hardware

- Digitized adiabatic ground-state preparation on Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps for second-order accuracy.
- Flattened the anneal schedule at the critical point, where a linear sweep races through the closing gap.

Entanglement and Matrix Product States: many-body systems slides

- Sequential SVD wavefunction decomposition measuring Schmidt spectra, entanglement entropy at every cut, and fidelity under truncation, relating bipartite entanglement structure to MPS compressibility.

TEACHING EXPERIENCE

Introduction to Quantum Computing, Columbia University · *TA* Jan 2026 – May 2026

Parallel and Concurrent Programming, Washington University in St. Louis · *TA* Jan 2025 – May 2025

Reverse Engineering and Malware Analysis, Washington University in St. Louis · *TA* Aug 2024 – Dec 2024

INDUSTRY EXPERIENCE

Smack Technologies · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulation for the core pipelines, integrating the knowledge graph to expand corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency, increasing corpus-generation throughput 64×.
- Validated simulator outputs against independently selected reference cases, bounding error to 0.55%, in an automated report that regenerates as the physics model changes.

Highnote · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, ingesting over 1 TB/day.

Mindtrip · *Software Engineer Intern* May 2023 – Jul 2023

- Built internal admin and secure-query tooling through Retool, saving roughly 250 engineering hours a month.

Bugcrowd · *Associate Application Security Engineer* May 2020 – Jul 2020

- Triaged and validated vulnerability submissions from a global researcher community for enterprise clients.

AWARDS AND COMPETITIONS

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street and Classiq challenge.
- **Qualified**, NYU Abu Dhabi Quantum Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

TECHNICAL SKILLS

Quantum software: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution and benchmarking.

Quantum methods: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT and block encoding, linear combination of unitaries, quantum walks, VQE and QAOA, Trotter decomposition.

Error correction: Rotated and XZZX surface codes, stabilizer formalism, minimum-weight perfect matching and Union-Find decoding, threshold estimation, randomized benchmarking, tomography, noise modeling.

Programming and tools: Python, C++, Rust, C, MATLAB, JavaScript, Go, Java · PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy · Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025 – Jun 2026). Ran weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms lecture notes, each paired with simulation checks.
- **Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.